University of British Columbia
Electrical and Computer Engineering
ELEC 291/292

Robot Assembly

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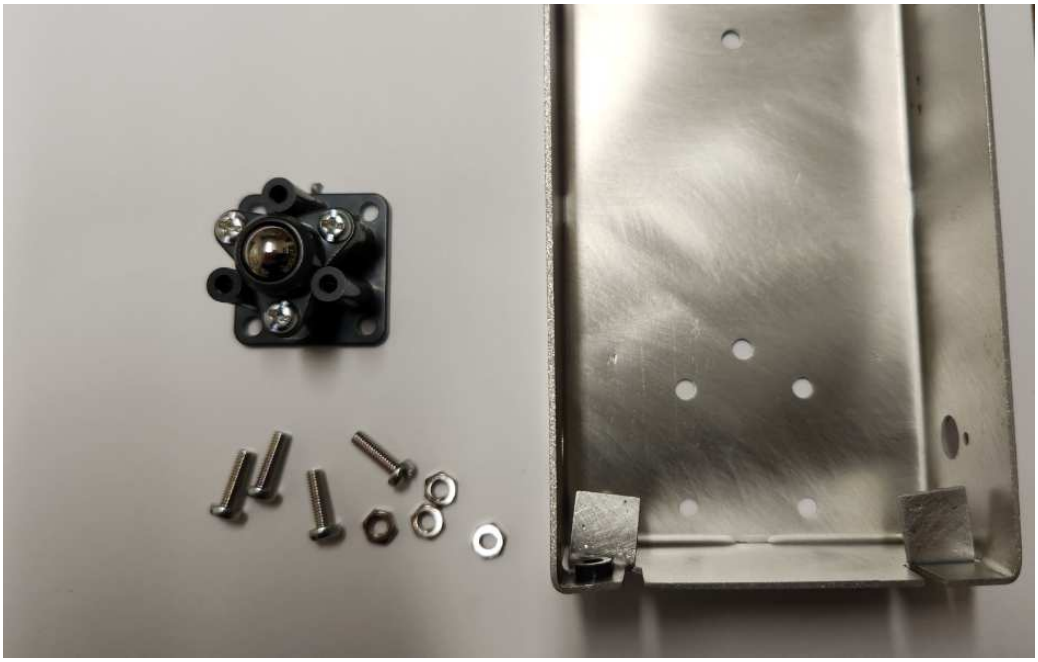
This document shows how to assemble the robot for project 2. The materials listed below are required. A picture of the materials from the project #2 parts kit is also shown below.

Description	Quantity
Ball caster kit	1
Geared motor	2
Servo wheels	2
Robot chassis	1
4-40/2-56 screw/nut kit	1
4 x AA battery holder	1
9V battery clip	1
DPDT Switch	1

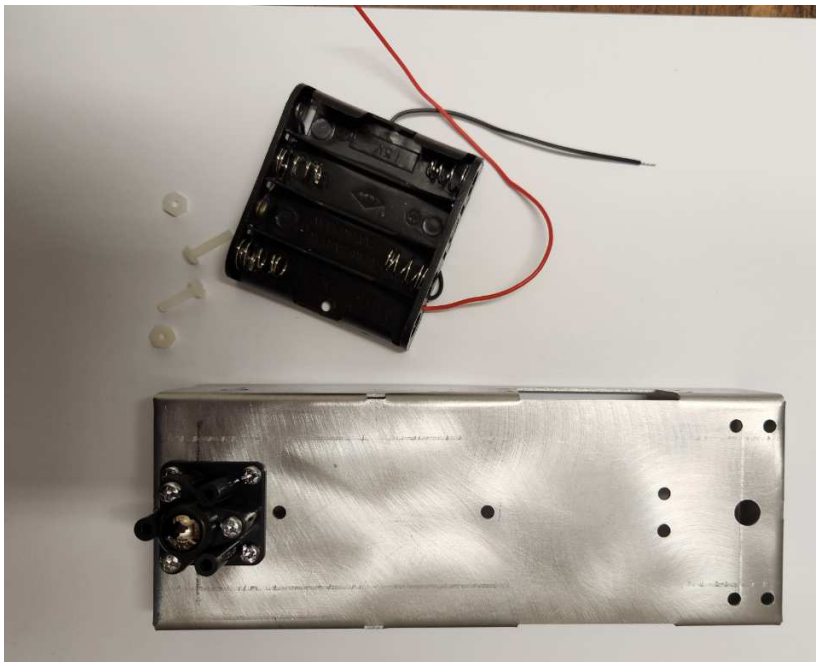
Steps

1) Follow the directions included with the caster wheel in order to assemble it. There are materials to assemble two caster wheels. We just need one. The caster wheel comes with machine screws to attach it to the robot chassis.

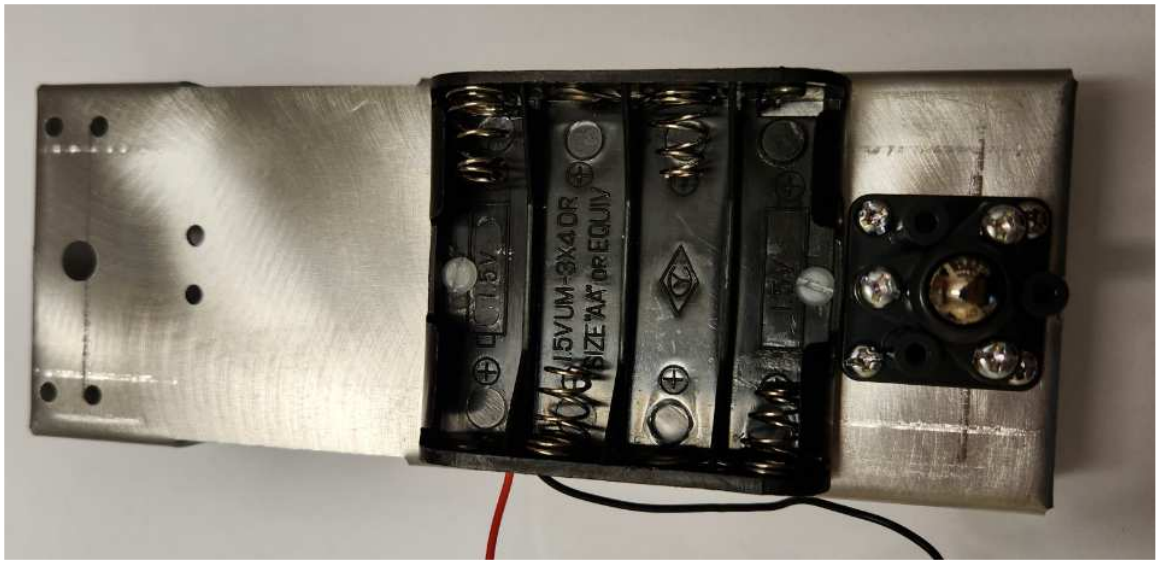


2) Attach the caster wheel to the robot chassis as shown in the picture bellow.

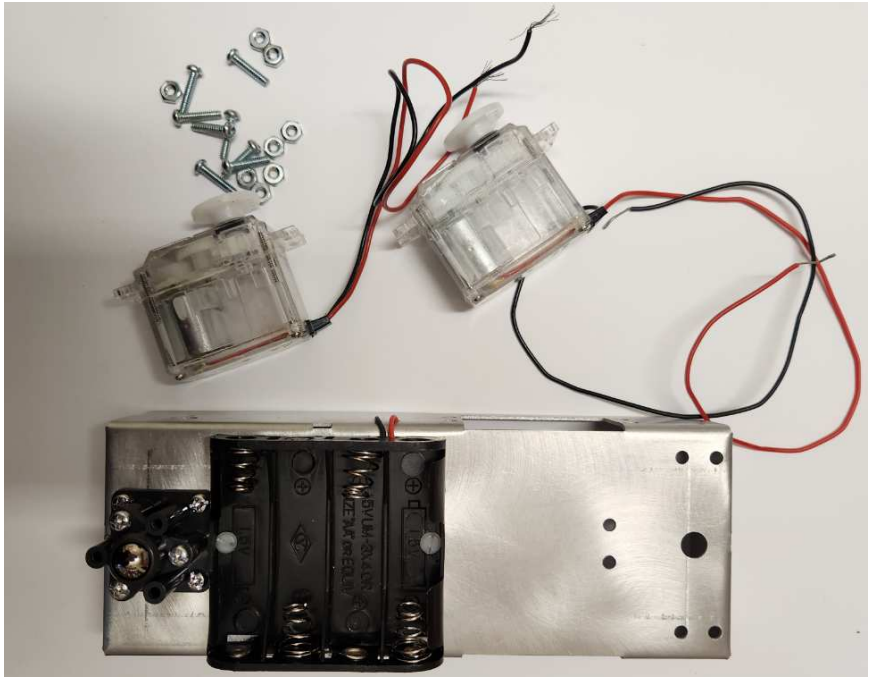
3) The next step is to attach the 4 x AA battery holder to the bottom of the robot. We will be using two 2-56 metal screws and nuts. In the picture bellow the screws are 4-40 nylon screws, which are hard to install and often break. Use the SMALLER 2-56 screws. 4-40 metal screws WILL SHORT CIRCUIT THE BATTERIES.



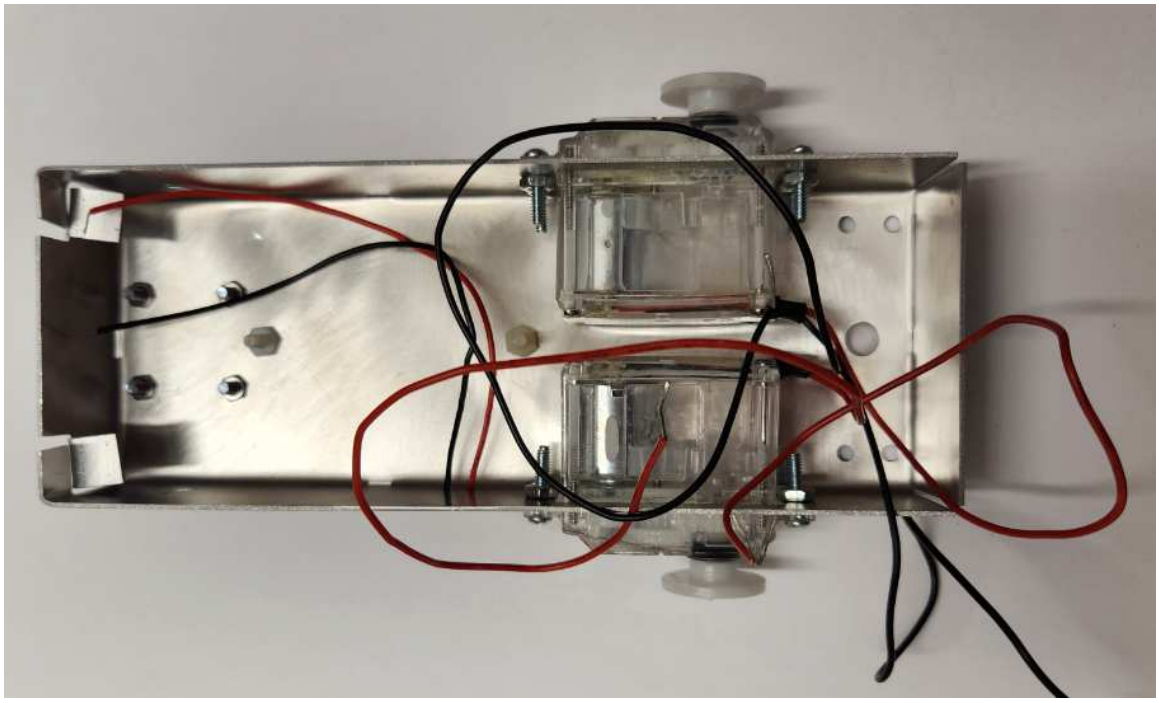
4) Attach the battery holder to the chassis using SMALL 2-56 screws.



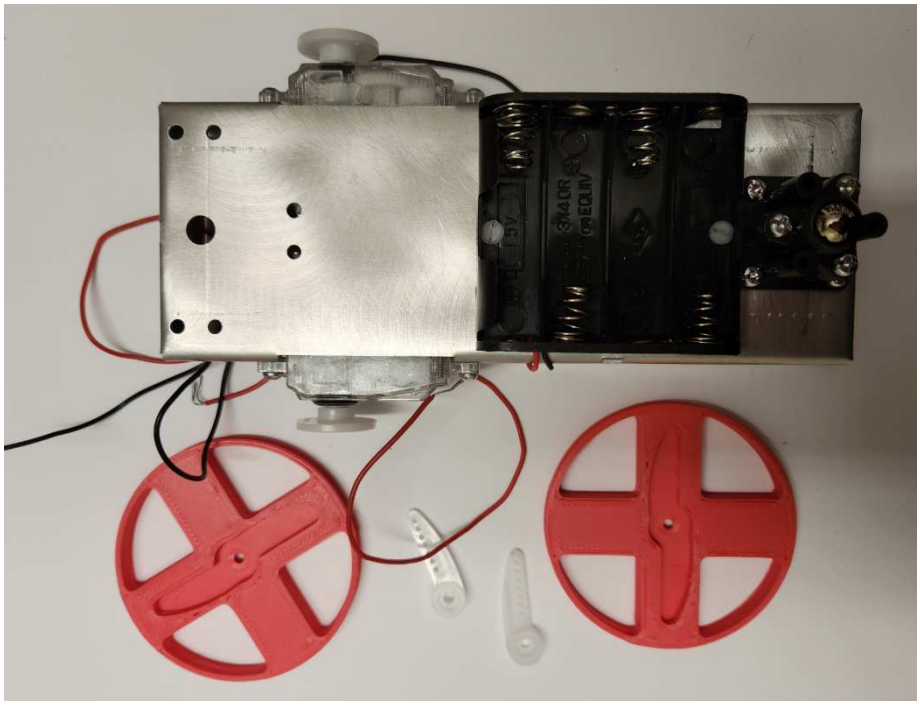
5) The next step is to attach the geared motors to the chassis. We need four 4-40 screws and nuts per motor.



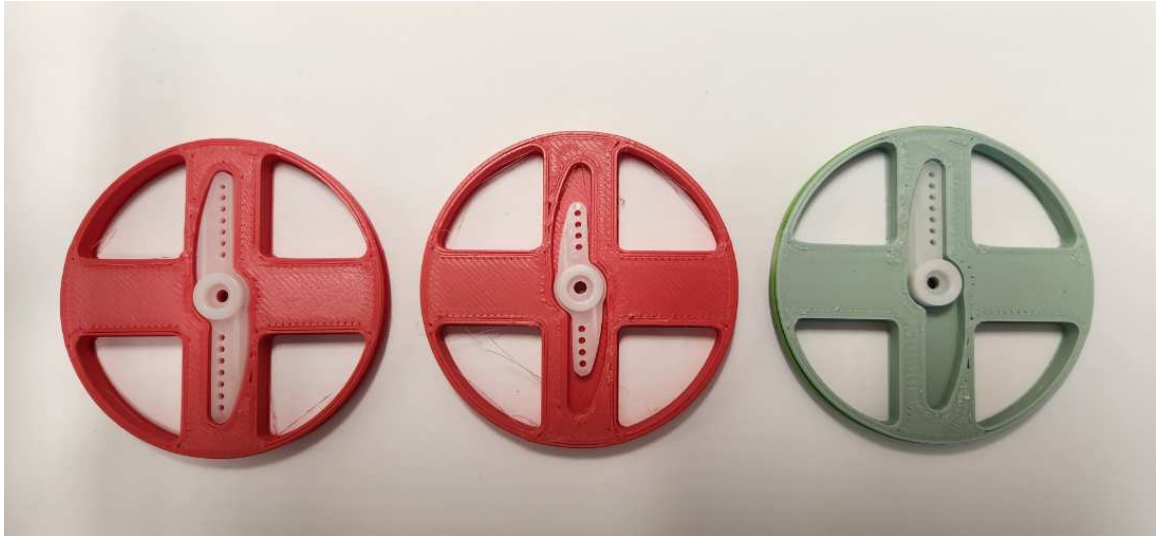
6) Install the geared motors as shown in the picture bellow.



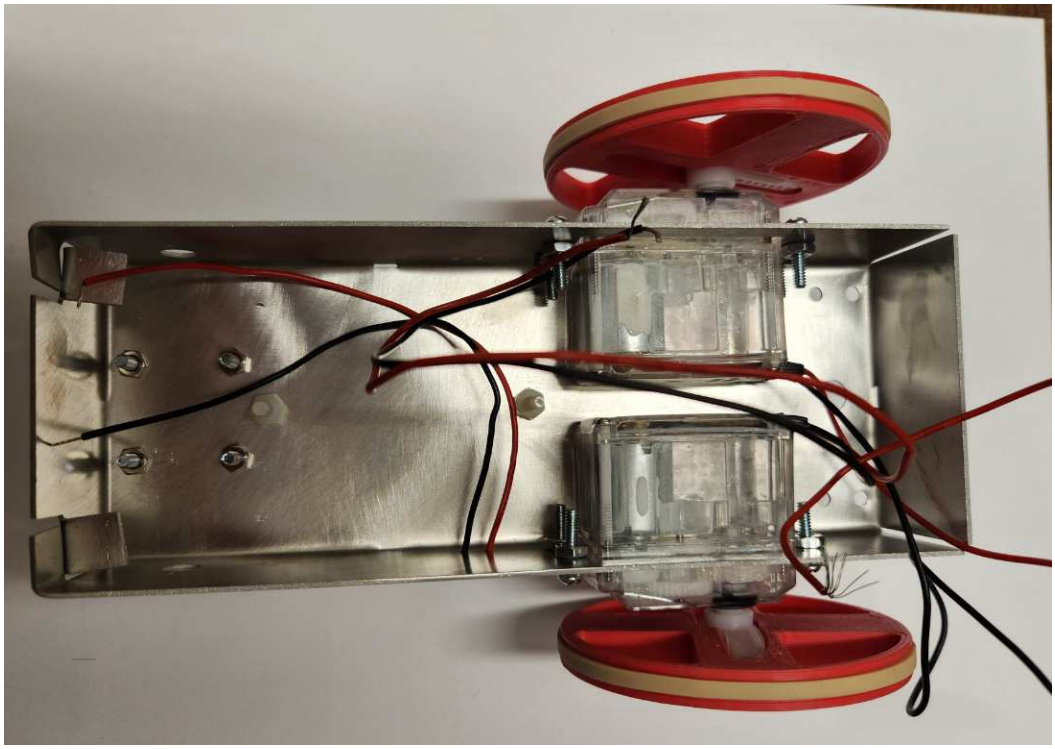
7) Now we can install the wheels. We need to remove the round plastic horns from the wheels first.



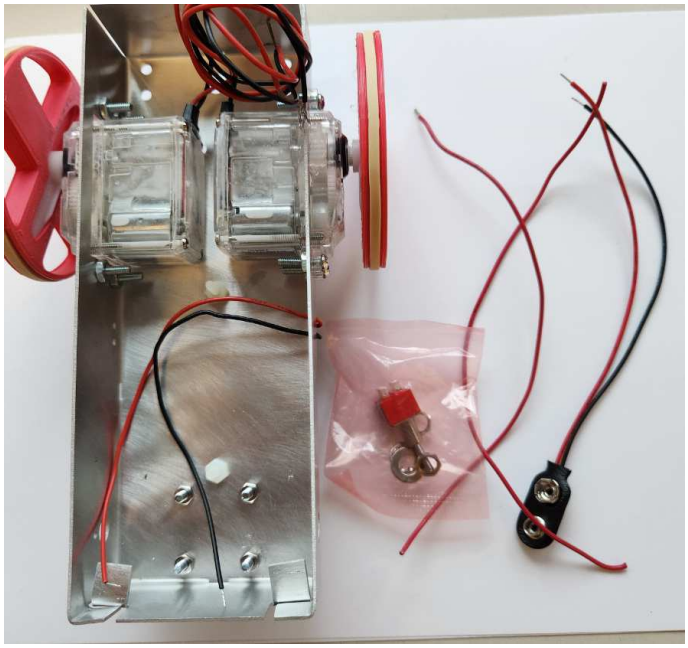
8) The wheels can accommodate three kinds of horns. The best one is the first in the picture below, but the other two work also:



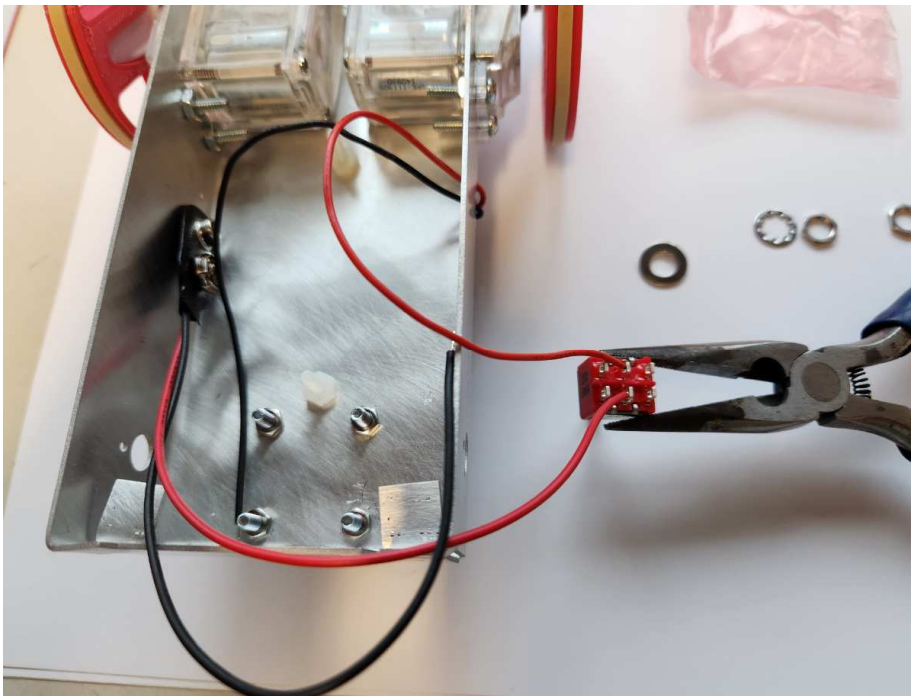
9) After removing the white circular motor horns that come with the servo motors, we install the wheels and secure them with the same black self threading screws that came with the motors.



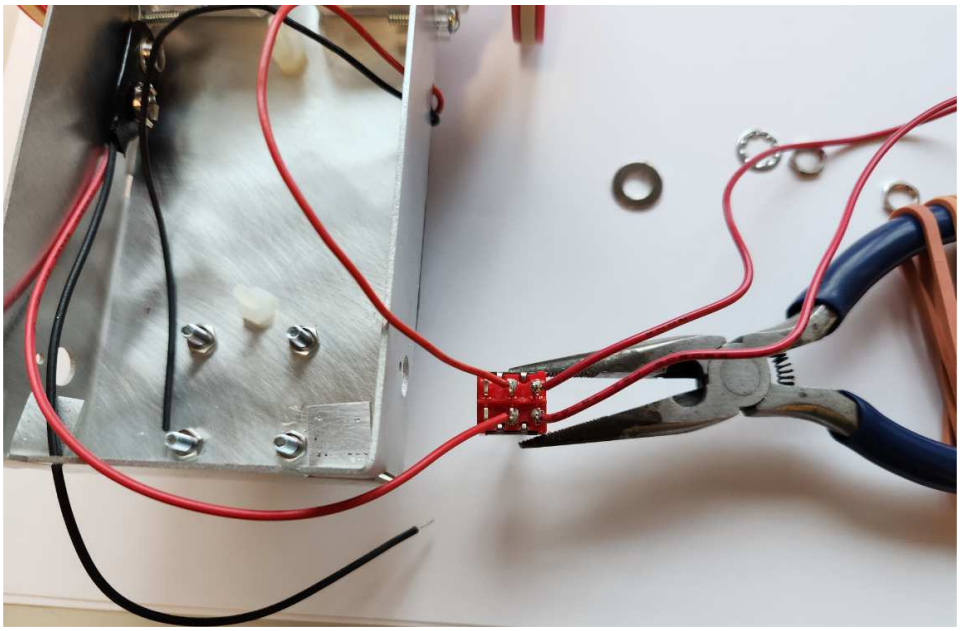
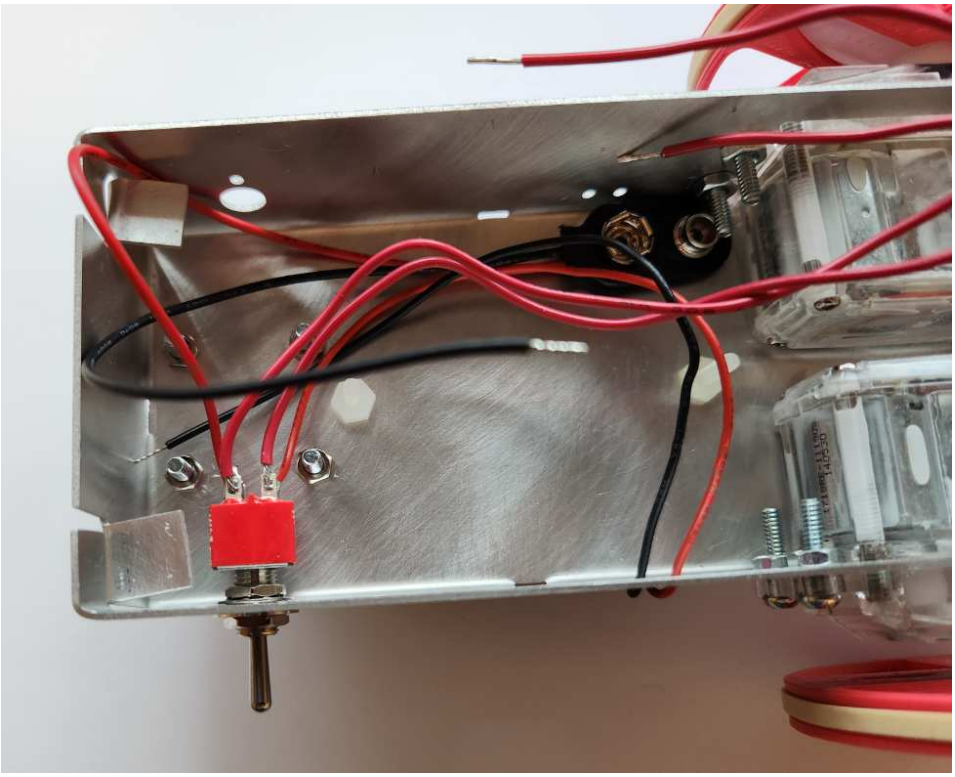
10) We can now install the double-pole/double-toggle DPDT switch allows us to turn the robot on/off.



11) Before installing the DPDT switch we need to solder some wires. Start by soldering the red wire of the 4xAA battery holder and the red wire of the 9V battery clip to the center poles of the switch. Note how the wires of the 4xAA battery holder come inside the robot chassis using the two small holes on the side.



12) Now solder two extra wires to one side of the extra poles of the switch.



13) Install the DPDT switch. The robot body is now complete.